Wen Quan

Academic CV for PhD Application in Computer Science and Technology
wenquan@student.usm.my | wenquan@hngm.edu.cn | +86 156-4900-6328 | +60 12-938-6817
Academic homepage: `wenquan0816.github.io/schoolar-site` | Portfolio: `age-friendly-community-map`
ORCID: 0009-0005-3975-8104

Application Objective

Application for the **Doctor of Philosophy in Computer Science and Technology** at Xiamen University Malaysia, with a proposed research focus on **knowledge-guided vision-language models, lightweight spatial intelligence, and AI-assisted assessment of age-friendly indoor environments**.

Profile

AI-assisted built environment researcher with a completed PhD in Interior Design from Universiti Sains Malaysia. My research connects age-friendly built environments, evidence-based design, public health resilience, spatial assessment, and applied AI methods. My published work has used LSTM-based model predictive control, natural language processing, SciBERT-assisted indicator extraction, contrastive learning, hyperbolic embedding, GANs, random forest regression, and multi-criteria assessment.

For the proposed PhD in Computer Science and Technology, my goal is to move from domain-specific AI applications toward computational methods for **knowledge-guided vision-language models and explainable spatial assessment systems** for age-friendly indoor environments.

Proposed PhD Research

Proposed title: A Knowledge-Guided Vision-Language Framework for Age-Friendly Indoor Space Assessment and Design Recommendation.

Research fit: Artificial Intelligence, Human-Computer Interaction, Software Technology, multimodal AI, knowledge-based systems, and applied computer vision for built environment assessment.

Education

- **PhD in Interior Design**, School of Housing, Building and Planning, Universiti Sains Malaysia, Malaysia, 2021–2026. Status: completed / viva passed. Research area: age-friendly built environments, AI-assisted assessment, interior and community environment evaluation.
- **Master of Engineering in Architecture**, Henan University of Technology, China, 2014–2017. Specialization: Architectural Design and Theory.
- **Bachelor of Engineering in Architecture**, Anyang Normal University, China, 2009–2014. Dual Degree Program: Bachelor of Education.

Research Interests

- Vision-language models for spatial assessment and design recommendation.
- AI-assisted age-friendly indoor and community environment evaluation.
- Knowledge-guided risk assessment, explainable AI, and decision support.
- Human-computer interaction for design and built environment applications.
- Natural language processing for design guidelines, policies, and expert knowledge extraction.

Computational and AI Experience

- LSTM-based model predictive control for building energy flexibility.
- NLP and SciBERT-assisted framework development for community resilience and age-friendly conversion studies.
- Contrastive learning and hyperbolic embedding for public health emergency resilience assessment.
- GANs and random forest regression for assessment and prediction modelling.
- Web-based interactive assessment demo using HTML, CSS, JavaScript, Leaflet, and ECharts.
- Multi-criteria assessment and expert validation: AHP, indicator weighting, framework development.

Selected Portfolio

China Age-Friendly Community Assessment Map
Live demo: `wenquan0816.github.io/age-friendly-community-map`
Repository: `github.com/WENQUAN0816/age-friendly-community-map`

This web-based demo visualizes multidimensional age-friendly community assessment results through an interactive map and dashboard. It demonstrates my ability to translate built environment research into computational scoring, spatial visualization, and user-facing prototype development.

Publications

WoS SCIE/SSCI Publications

1. Quan Wen; Mazran Ismail; Muhammad Hafeez Abdul Nasir (2026). A LSTM-based model predictive control method for unlocking the potential of building energy flexibility in Malaysian commercial buildings. *Scientific Reports*. DOI: `10.1038/s41598-026-53501-8`. WoS SCIE Q1; first author.
2. Quan Wen; Mazran Ismail; Muhammad Hafeez Abdul Nasir; Yanting Wu; Chaojiang Hu (2026). AFLE-HIV: developing an age-friendly living environment assessment framework for elderly HIV-infected individuals in China. *BMC Public Health*. DOI: `10.1186/s12889-026-27748-9`. WoS SCIE Q1; first author.
3. Quan Wen; Mazran Ismail; Muhammad Hafeez Abdul Nasir (2026). Transforming vacant rural school buildings into age-friendly facilities under China's aging and low-fertility context: A natural language processing-based evaluation framework. *BMC Geriatrics*. DOI: `10.1186/s12877-025-06772-1`. WoS SCIE Q1; first author.
4. Quan Wen; Mazran Ismail; Muhammad Hafeez Abdul Nasir (2025). A community public health emergency resilience assessment framework based on contrastive learning and hyperbolic embedding. *Frontiers in Public Health*. DOI: `10.3389/fpubh.2025.1651331`. WoS SSCI Q1; first author.
5. Yanli Han; Quan Wen (2026). Emotion Recognition in Literary Texts via an Attention-Guided Dual-Stream Gated Fusion Network. *IEEE Access*. DOI: `10.1109/ACCESS.2026.3686065`. WoS SCIE; second author; corresponding author.
6. Wenjing Xu; Mazran Ismail; Syafizal Shahruddin; Wen Quan; Yingrui Li (2025). Exploring tourists' intentions to adopt augmented reality in cultural heritage museums: Insights from a modified technology acceptance model. *SAGE Open*. DOI: `10.1177/21582440251339936`. WoS SSCI Q1; co-author.

WoS ESCI and Scopus-Indexed Publications

1. Jia Sheng Zhou; Mohd Jaki Bin Mamat; Quan Wen (2025). Analyzing the traditional rural cultural heritage of Lingnan region in China as the foundation for protection and development. *International Journal of Conservation Science*. DOI: `10.36868/ijcs.2025.01.06`. WoS ESCI Q1; co-author.

2. Quan Wen; Mazran Ismail; Muhammad Hafeez Abdul Nasir (2024). Evaluating the residential environment of traditional settlements in northern Jiangxi, China: A multi-dimensional framework. *Journal of Construction in Developing Countries*. DOI: `10.21315/jcdc.2024.29.s1.11`. WoS ESCI Q4; first author.

3. Zhang Jing; Hasnanywati Hassan; Zalena Abdul Aziz; Khoo Terh Jing; Wen Quan (2025). Exploring the factors for the use of building information modelling (BIM) in historic buildings. *Journal of Construction in Developing Countries*. DOI: `10.21315/jcdc.2025.30.s2.14`. WoS ESCI; co-author.

4. Quan Wen; Mazran Ismail; Muhammad Hafeez Abdul Nasir; Jing Zhao; Zhang Jing (2025). Assessment and prediction model for community resilience in China based on GANs and random forest regression. *Planning Malaysia*. DOI: `10.21837/pm.v23i38.1823`. Scopus-indexed; first author.

5. Quan Wen; Mazran Ismail; Muhammad Hafeez Abdul Nasir; Jing Zhao; Zhang Jing (2025). Developing a comprehensive framework for assessing community resilience to public health emergencies in Henan Province: Insights from SciBERT analysis. *Planning Malaysia*. DOI: `10.21837/pm.v23i38.1822`. Scopus-indexed; first author.

Research Projects

- **Comprehensive Aging-in-Place Renovation Framework for Legacy Residential Communities in Henan Province** (2022). Henan Provincial Social Science Research Fund, Project No. SKL-2022-608. Role: Principal Investigator.
- **Innovative Universal Design Strategies for New Community Development in Henan Province** (2018). Henan Provincial Social Science Research Fund, Project No. SKL-2018-490. First Prize. Role: Principal Investigator.
- **Interior Environment Design Strategies for Hospital Outpatient Departments in the Era of Telemedicine** (2020). Henan Provincial Social Science Research Fund, Project No. SKL-2020-1566. Role: Principal Investigator.
- **Integrated Healthcare-Housing Model for Community-Based Senior Living Facilities in Zhengzhou** (2018). Zhengzhou Federation of Social Sciences Research Project, Project No. JX20190146. Role: Principal Investigator.
- **Adaptive Renovation Strategies for Existing Long-term Care Facilities under COVID-19 Safety Protocols** (2021). Henan Provincial Department of Education Research Fund, Project No. 2021-ZDJH-433. Role: Principal Investigator.

Teaching and Professional Experience

- Lecturer, Henan Industry and Trade Vocational College, China, 2021–2023.
- Part-time Lecturer, Henan University of Animal Husbandry and Economy, China, 2019–2021.
- Lecturer, Easton International Art College, Zhengzhou University of Light Industry, China, 2017–2021.
- Research Associate, Green Building Research Institute, Henan Academy of Building Research, China, 2017.

Professional Service and Certifications

- Journal Reviewer, *SAGE Open*.
- Academic Committee Member and Senior Project Review Expert, Zhongren Art Museum.
- Higher Education Teaching Qualification Certificate.
- Lecturer Certificate, Zhengzhou University of Light Industry.
- National Computer Proficiency Examination Certification.
- Network Engineering Professional Certification.

Languages

- Chinese Mandarin: Native.
- English: Advanced academic and professional proficiency; TOEFL 86/120.
- Malay: Conversational proficiency.

Referee

Dr. Mazran bin Ismail, Senior Lecturer, Architecture, School of Housing, Building and Planning, Universiti Sains Malaysia. Email: mazran@usm.my. Relationship: PhD supervisor.